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Navbea RTLS Solutions

Real-Time Location Solutions

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KODAR

2013

Kodar Organization

2014

RTLS R&D Project

2015

US Export BLE Sensor Production

2016

Technological Product Experience Certificate for Developing IoT-Based Efficiency and OSH Applications

2017

Machine Monitoring and Remote Management Solutions

2018

NASA Space Center Houston Solution Provider

2019

Artificial Intelligence Computer Vision Deep Learning Started

2020

Implementation of RTLS Systems in 4 Major Theme Parks and 4 Major Thermal Power Plants in Azerbaijan

NAVBEA

2021

Artificial Intelligence Occupational Safety Solutions Implemented

2022

European office opened in Amsterdam, Netherlands

2023

We started working on international projects.

2024

AI-Enabled Asset Digitalization and IoT Integration Framework

2025

AI-Enabled Hybrid Real-Time Logistics Network (RTLS) Architecture







YATAĞAN
TERMİK ENERJİ

Yatağan Thermal Power Plant

Yatağan Thermal Power Plant is Turkey's fifth largest thermal power plant, located 3 km from the town of Yatağan, which is 26 km north of Muğla, covering an area of 1,163 m².

Navbea Solutions Used;

- 3800 Personnel location information,
- Time and Attendance, Shift management,
- Emergency management,
- Inactivity detection,
- Panic button,
- Alarm in case of unauthorized area presence,
- SMS and Email notifications
- Facility efficiency monitoring and analysis
- Micro-location based task tracking
- Workflow tracking



Yeniköy Thermal Power Plant

Yeniköy-Kemerköy Electricity Generation and Trade Inc., with a total installed capacity of 1050 MW, supplying approximately 2% of Turkey's electricity needs, is an important economic factor in the Aegean Region, with 420 MW from 2 units at Yeniköy Thermal Power Plant and 630 MW from 3 units at Kemerköy Thermal Power Plant.

Navbea Solutions Used;

Personnel Device: 767
Guest Device: 108
Gateway: 70
Access Control Device: 23

- Time and Attendance, Shift management,
- Emergency management,
- Panic button,
- Alarm in case of unauthorized area presence,
- SMS and Email notifications.
- Micro-location based task tracking



Rönesans Sancaktepe Construction Site

The Sancaktepe City Hospital project, carried out by Rönesans Holding, is one of Turkey's largest health investments, conducted with thousands of workers across a large construction site. Navbea RTLS infrastructure has been established for the safety of personnel working on-site, operational efficiency, and emergency management. The system allows real-time tracking of personnel's location both indoors and outdoors.

Navbea Solutions Used;

1000 Lorawan Personnel Cards
200 Navbea BLE Long Range Sensors
4 Lorawan Gateways

- Real-time personnel location tracking
- Indoor and outdoor positioning
- Digital monitoring of construction workers
- Emergency management and rapid personnel counting
- Detection of unauthorized area entries
- Facility operation and efficiency analysis



Rönesans – Office AI Personnel Positioning

In Rönesans Holding offices, the Navbea AI-supported positioning system is used to analyze employees' location information within the office. The system enhances location accuracy by analyzing sensor data with artificial intelligence algorithms and provides insights into the usage efficiency of office spaces.

Navbea Solutions Used;

2000 Personnel Cards,
140 IP54 Readers,
20 IP68 Readers

- In-office personnel positioning
- AI-supported location verification algorithms
- Space usage analysis
- Office efficiency reports
- Workspace optimization



Rönesans AVM – Personnel Location and Task Tracking

In Rönesans AVM projects, the Navbea RTLS system is used to track the locations and tasks of personnel working in the shopping mall. The system facilitates operational management by analyzing whether personnel visit their task areas and their working durations.

Navbea Solutions Used;

12 shopping malls, 1400 Personnel Cards, 1985 Readers

- Shopping mall personnel location tracking
- Time and Attendance tracking
- Task point control
- Analysis of personnel visits to task locations
- Operational efficiency analysis
- Live personnel monitoring



Migros

Migros is one of Turkey's largest retail chains, with a wide product inventory including food, cleaning, cosmetics, household goods, and electronics. It employs tens of thousands of workers across thousands of stores, organized among sales, logistics, warehouse, customer service, and administrative units. Migros also operates with modern retail technologies such as supply chain management and dynamic inventory management.

Navbea Solutions Used;

Navbea T1 Multi-Purpose Tracking Sensor (2000 units)
Navbea Asset Sensor (3000 units)
Total of 5000 inventory sensors
Smart inventory tracking



Çimsa – Production Facility Personnel Tracking

The Navbea RTLS system has been established in Çimsa cement production facilities for tracking employee locations, ensuring occupational safety, and analyzing production efficiency. The system analyzes employee movements and working hours in the production area to enhance operational efficiency.

Navbea Solutions Used;

Personnel Devices: 700
Gateway: 20

- Personnel location tracking
- Time and attendance and shift management
- Occupational safety monitoring
- Movement analysis in the production area
- Personnel efficiency reports
- Emergency management.





Medcem

Medcem Mining is a company located in the Silifke district of Mersin, home to one of Turkey's largest integrated cement plants. Operation area is approximately 500,000 m² and has a daily clinker production capacity of 9,000 tons. It employs over 500 personnel. Medcem's operations encompass a wide range of activities including cement production, mining, logistics, and export-oriented trade. With its port infrastructure, it serves both the domestic and international markets.

Navbea Solutions Used;

100 Personnel Cards, 25 IP54 Readers, 25 IP68 Gateways, 4 Card Readers

- Personnel location information,
- Time and Attendance Management, Shift Management,
- Emergency management,
- Inactivity detection,
- Panic button,
- Alarm for unauthorized area presence,
- SMS and Email notifications,
- Facility efficiency monitoring and analysis,
- Micro-location based task tracking,
- Workflow tracking.





Çalık Energy – Libya Personnel Tracking System

The Navbea RTLS system has been established for the safety and operational management of personnel working on Çalık Energy projects in Libya. The system allows for the monitoring of field workers' locations, digitizing safety and efficiency processes.

Navbea Solutions Used;

Personnel Devices: 1250
Gateway: 30

- Personnel location tracking
- Time and attendance and shift management
- Occupational safety monitoring
- Emergency management
- Control of unauthorized areas
- Operational efficiency analysis.







Limak – AI Telemetry Vehicle Tracking

Construction machinery, trucks, and field equipment used in Limak projects are monitored with the Navbea Telemetry platform. Sensor and telemetry data collected from vehicles are analyzed with artificial intelligence algorithms to provide operational teams with action recommendations regarding vehicle operational status, performance, and potential failure risks.

Navbea Solutions Used;

- Construction machinery and vehicle telemetry tracking
- Real-time analysis of vehicle sensor data
- AI-supported vehicle behavior analysis
- Failure and risk prediction algorithms
- Operational efficiency analysis
- Vehicle performance reports



BRANDWEER



Dutch Fire Brigade – Digital Twin

The Navbea Digital Twin solution developed for Dutch fire brigade teams is designed to analyze firefighters' movements inside buildings during fire interventions. Movement data collected with IMU sensors is analyzed in real-time and transformed into a digital operational map. This system allows command centers to monitor firefighters' locations, movements, and health status live.

Navbea Solutions Used;

- Movement analysis with IMU sensors
- Fire operation digital twin system
- Real-time indoor location analysis
- Tracking of fire personnel movements
- Live operational map
- Health data and operational safety monitoring



E-Garage – Machine and Personnel Efficiency Tracking

The Navbea RTLS system has been established in E-Garage facilities to enhance the efficiency of production and operational processes. The system analyzes the location and working hours of personnel working at the machine, providing detailed efficiency reports on production processes.

Navbea Solutions Used;

30 Personnel Cards, 20 Readers

- Personnel tracking at the machine
- Analysis of personnel working hours
- Measurement of production efficiency
- Workflow analysis
- Operational performance reports.





Unilever – Forklift Authorization System

An IoT-based authorization system has been established at Unilever facilities for the safe use of forklift vehicles. The system controls forklift usage according to operator permissions and monitors safety rules such as seatbelt usage.

Navbea Solutions Used;

- Forklift operator authorization
- Seatbelt control system
- Prevention of unauthorized use
- Vehicle usage records
- Occupational safety management.







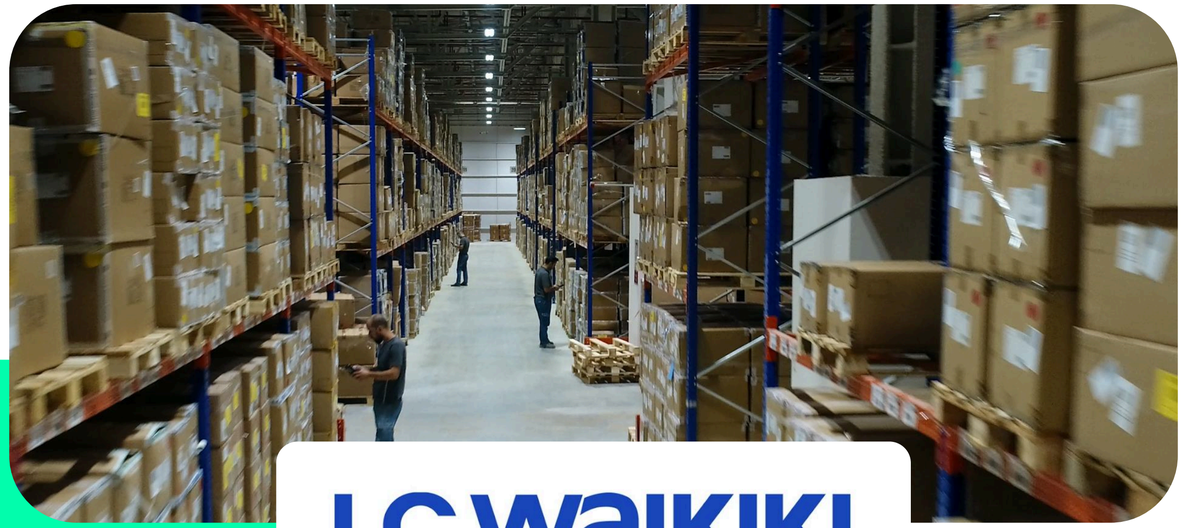
Kinderland Bakü

With a total area of 5000 m², Kinderland resembles a real city with its streets, shops, transportation system, and unique currency. This city is aimed at children aged 4-14 (and infants aged 2-4 in the region). Kinderland combines entertainment and education with over 60 real-life activities.

Navbea Solutions Used;

Navbea Wristband: 5000
Gateway: 47

The world's first RTLS Child Safety and Experience Analysis Project
- Real-time location information for children,
- Navbea mobile application for parent-child tracking,
- Emergency management,
- Spending information.







LC Waikiki – AI Camera Analysis System

The Navbea AI camera analysis system used in LC Waikiki operations analyzes camera images in real-time to detect anomalies. The system learns normal operational behaviors through image processing algorithms and automatically detects situations that deviate from these behaviors, generating alarms.

Navbea Solutions Used;

Number of Cameras: 4100
Number of Use Cases: 80

- AI-based video analysis
- Anomaly detection
- Behavior analysis
- Automatic alarm generation
- Security and operation monitoring.



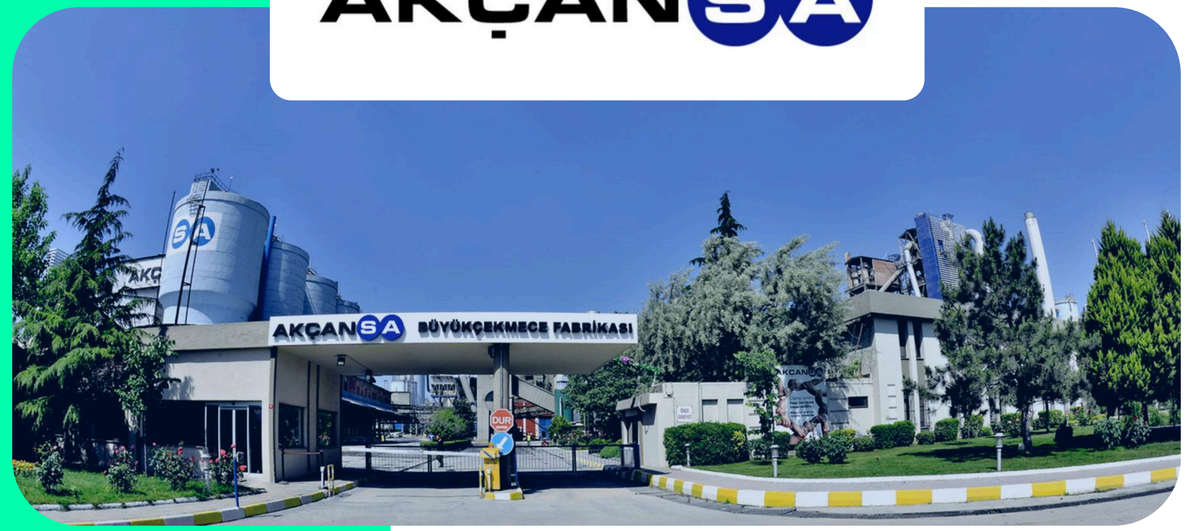
Eren Energy

Eren Energy is an energy company located in Zonguldak Muslu, owning one of Turkey's largest private electricity generation facilities. With a total installed capacity of 2,790 MW, it produces approximately 6% of the country's electricity. The company, headquartered in Istanbul Ataşehir, operates with a workforce of around 1,000 people; it operates thermal, biomass, and waste incineration plants. Additionally, it manages logistics for coal and by-products through its own port.

Navbea Solutions Used;

- Lone worker safety,
- Panic button,
- Inactivity detection,
- Alarm for unauthorized area presence,
- Voice alert system to the security department,
- SMS and Email notifications.





Akçansa – Personnel Tracking System

In Akçansa cement production facilities, the Navbea RTLS infrastructure has been used to digitalize employee location tracking and time and attendance processes. The system was implemented without any changes to the existing facility infrastructure using battery-powered sensors.

Navbea Solutions Used;

- Personnel location tracking
- Contactless time and attendance tracking
- Quick installation with battery-powered sensor infrastructure
- Monitoring of personnel safety
- Analysis of working hours
- Operational efficiency reports.



Başakşehir Çam Sakura Hospital – Indoor Navigation

The Navbea indoor navigation solution has been used in Başakşehir Çam Sakura City Hospital to help visitors easily find their way within the hospital. The system enables route creation within the hospital through a digital map and mobile application.

Navbea Solutions Used;

- Indoor navigation system
- Digital mapping within the hospital
- Visitor guidance system
- Route creation and directions
- Improving patient and visitor experience.



Alarko – Personnel Gamification Project

A gamification-based efficiency system has been developed through the Navbea platform to enhance the performance of personnel working in Alarko production facilities. By analyzing the tasks completed and working hours through screens in the production area, motivation and performance have been increased.

Navbea Solutions Used;

- Personnel location tracking
- Task completion analysis
- Measurement of working hours
- Gamification-based performance system
- Production efficiency analysis.



BesiktasShipyard





Beşiktaş Shipyard – Forklift and Personnel Safety

An IoT-based safety system has been established at Beşiktaş Shipyard to enhance forklift and personnel safety. The interaction between the wristband worn by personnel and the readers on the forklift is analyzed to prevent collision risks.

Navbea Solutions Used;

- Forklift-personnel interaction detection
- Collision prevention system
- Audible warning system
- Vibrating safety alert
- Operational safety monitoring.

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